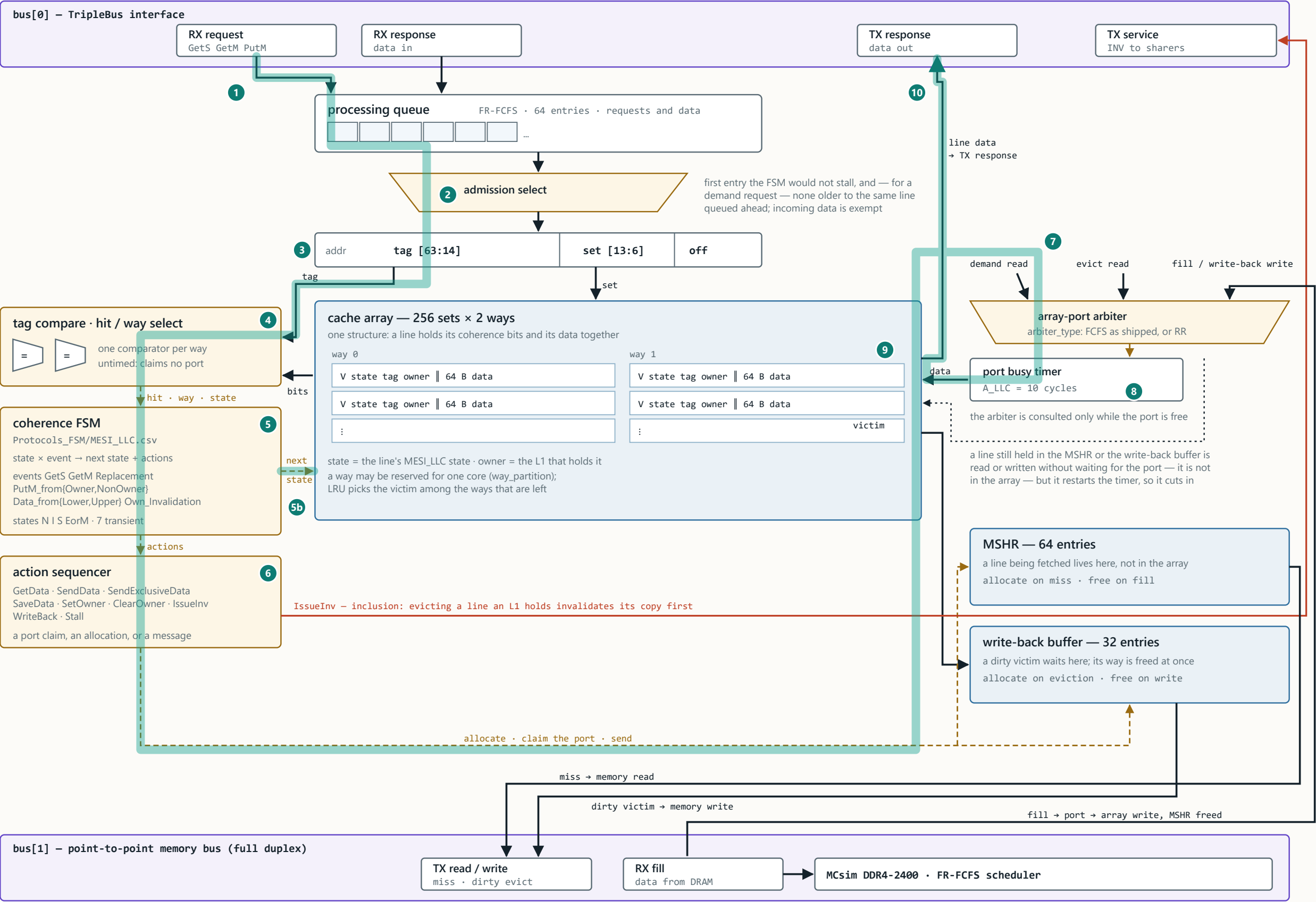


Octopus LLC datapath — the path of a write hit

CacheController_End2End + CacheDataHandler_COTS · 32 KB, 2-way, 64 B lines · MESI_LLC coherence FSM · one timed data port

Highlighted: a GetM for a line the LLC holds valid and no L1 owns (I or S). Same path as a read hit; the FSM answers with plain data and rewrites the owner bits — the untimed branch — and the line goes to EorM. Sharers drop their copies by snooping the GetM; the LLC sends no INV.



— data path (timed) register-resident access, bypasses the port - - - - - control (untimed) — back-invalidation (inclusion)

□ storage □ control logic □ interconnect interface the Logger and the raw trace observe: queue arrival · admission · port claim · data emitted

write hit — 1 arrives · 2 admitted · 3 decoded · 4 tag hit (untimed) · 5 FSM: SendData, SetOwner → EorM · 5b owner bits written into the line (untimed) · 6 sequencer claims the port · 7 arbiter grants · 8 busy timer · 9 array read · 10 plain data out on TX response